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ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

Department of Cyber Security

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III Year /V Sem

CS23533-FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

Review 1– 23.09.2026

Introduction

- Cybersecurity incidents such as Brute Force, Port Scanning and Denial-of-Service attacks can generate a large number of security alerts.
- Manually analyzing and prioritizing these incidents is time-consuming and error-prone.
- The proposed system uses Artificial Intelligence techniques and rule-based reasoning to analyze security incidents.
- It classifies incidents, determines their severity and priority, and recommends suitable response actions.
- A web-based dashboard provides a centralized view of incident details, AI analysis and response recommendations.

Domain: Cybersecurity + Artificial Intelligence

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Problem Statement

- Organizations receive a large volume of cybersecurity alerts from different sources.
- Manual investigation of every alert requires significant time and human effort.
- It is difficult to quickly determine which incidents are critical and require immediate action.
- Traditional systems may record incidents but provide limited intelligent analysis and response recommendations.
- Therefore, there is a need for an intelligent system that can analyze, prioritize and recommend responses automatically.

Need for the Solution

Automate incident analysis → Prioritize threats → Recommend appropriate response

Motivation

- Increasing frequency and complexity of cybersecurity attacks.
- Security teams need faster identification of high-priority incidents.
- Artificial Intelligence can assist in decision-making and automated reasoning.
- Reduces the dependency on completely manual incident analysis.
- Provides a practical application of AI concepts such as:
  - Intelligent Agents
  - Heuristic Evaluation
  - Constraint Satisfaction
  - Rule-Based Reasoning
  - Forward Chaining

Practical Significance:
Helps security teams respond faster and more systematically to potential cyber threats.

Objectives

1. To develop a centralized system for collecting and managing cybersecurity incidents.
2. To classify security incidents using rule-based AI techniques.
3. To determine the severity and priority of detected incidents using heuristic evaluation.
4. To recommend suitable response actions based on incident conditions.
5. To provide a web-based dashboard for monitoring incidents and AI-generated recommendations.

Literature Survey

S.No.	Author & Year	Method / Technique	Key Findings	Limitations
1	Buczak & Guven, 2016	Machine Learning for Cybersecurity	ML can support intrusion detection and cybersecurity analytics.	Requires suitable datasets and feature engineering.
2	Sommer & Paxson, 2010	Machine Learning for Network Security	Discussed challenges in applying ML to real-world intrusion detection.	Real-world network data can be difficult to model accurately.
3	Khraisat et al., 2019	Intrusion Detection Techniques	Reviewed different intrusion detection approaches and their challenges.	No single technique performs optimally for all attack types.

Existing System

Existing approaches include:

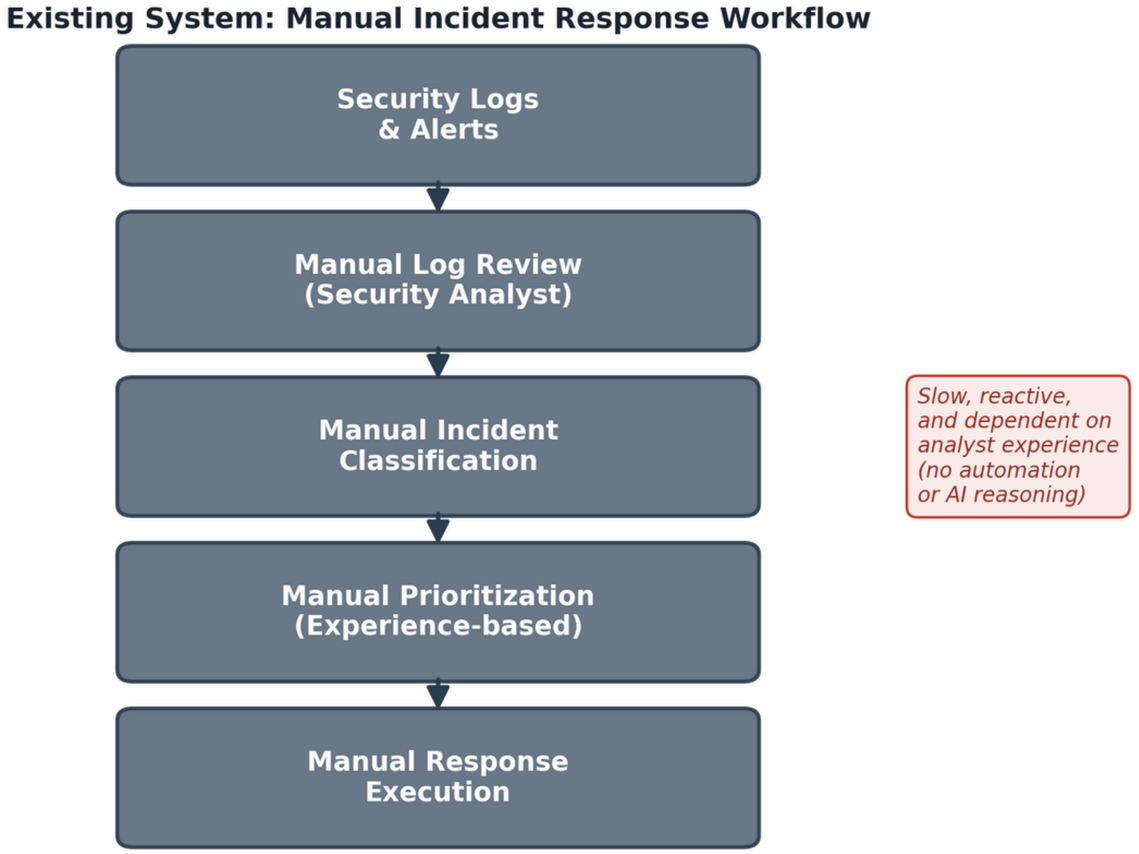
- Manual security incident investigation.
- Traditional rule/signature-based intrusion detection.
- Security monitoring systems that generate alerts.
- Separate tools for detection, analysis and response.

Advantages:

- Simple to implement.
- Can detect known attack patterns.
- Useful for basic security monitoring.
- Established techniques are available.

Limitations:

- Large number of alerts can overwhelm security teams.
- Manual analysis is time-consuming.
- Limited intelligent prioritization.
- Response decisions may depend heavily on human expertise.
- Detection and response may be handled by separate systems.



Proposed System

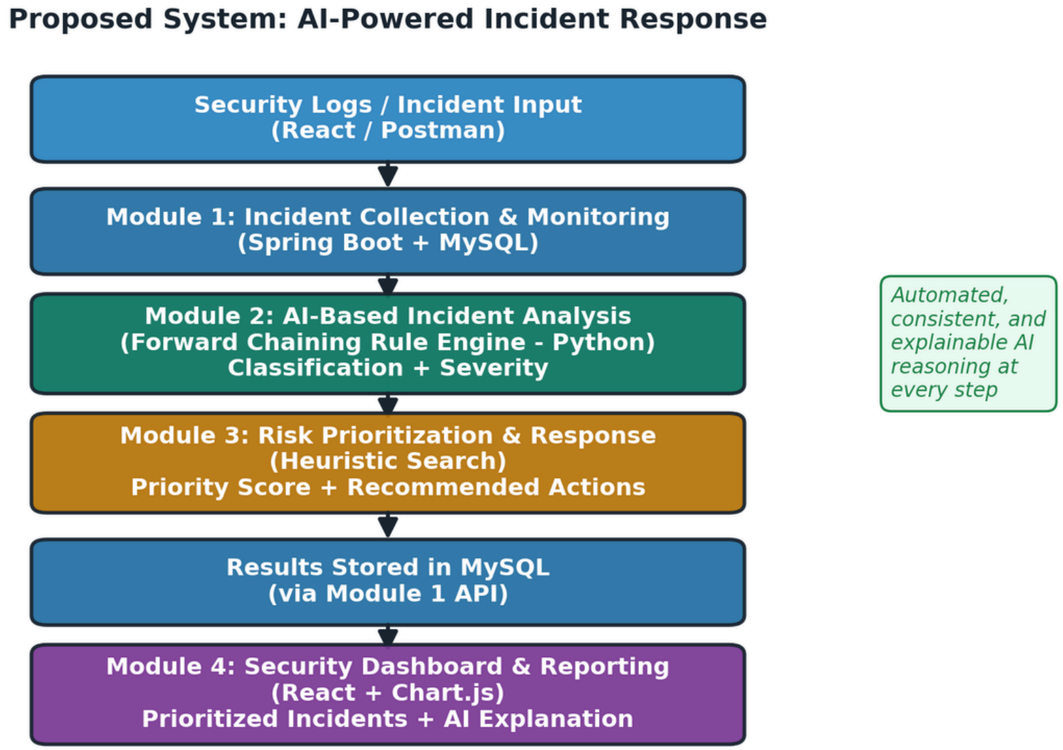
Proposed System
The proposed system is an AI-powered web application that analyzes cybersecurity incidents and recommends appropriate response actions.

Key Features

- Centralized incident management.
- Rule-based expert system.
- Forward-chaining inference.
- Heuristic-based incident prioritization.
- Constraint-based response selection.
- AI-generated severity and response recommendations.
- Incident history and monitoring dashboard.
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How It Improves Existing Systems
Incident Data → AI Analysis → Threat Classification → Priority → Response Recommendation
Expected Outcome

Faster and more systematic cybersecurity incident analysis with AI-assisted response decisions.



System Architecture / Block Diagram



Methodology

1. Incident Data Collection

2. Data Preprocessing

3. Feature Processing

4. AI-Based Analysis

5. Heuristic Evaluation

6. Response Recommendation

Technologies Used

Programming Language: [Java, JavaScript, HTML, CSS, SQL]
Framework / Library: [Spring Boot]
Database: [MySQL]
Tools / IDE: [VsCode, MySQL Workbench, Postman, Git / GitHub]
Hardware: [Standard computer/laptop, Minimum 8 GB RAM recommended]

Module Description

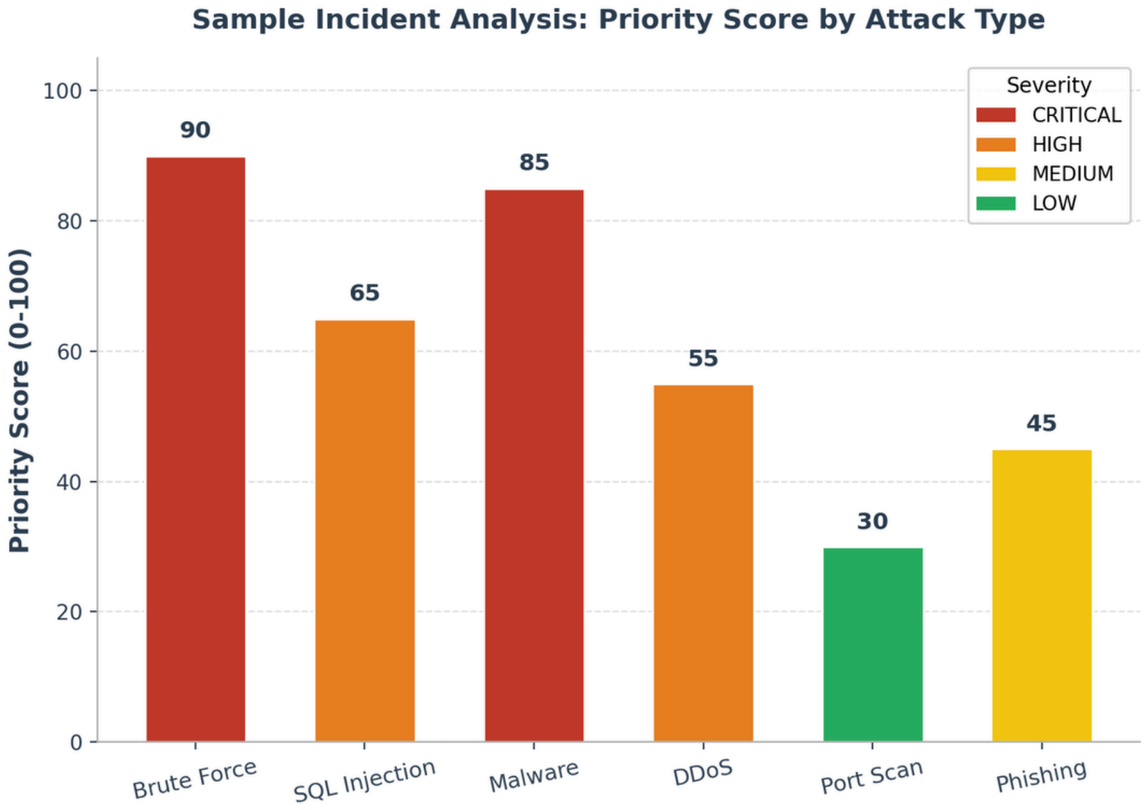
Module 1 – Incident Management & Backend

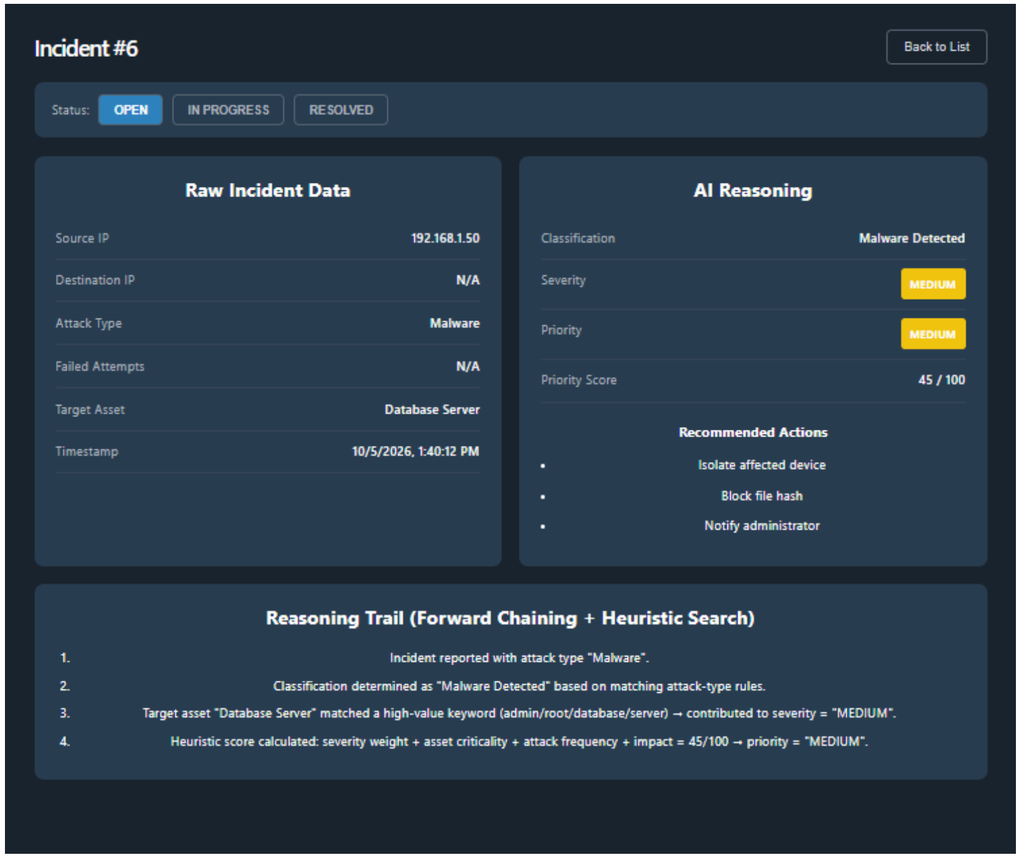
Module 2 – AI - Based Incident Analysis

Module 3 – Intelligent Prioritization & Response

Module 4 – Dashboard & Integration

Results


Sample Incident Analysis: Priority Score by Attack Type
Priority Score (0-100)
100
80
60
40
20
0
90
65
85
55
30
45
Brute Force
SQL Injection
Malware
DDoS
Port Scan
Phishing
Severity
CRITICAL
HIGH
MEDIUM
LOW


Incident #6
Back to List
Status:
OPEN
IN PROGRESS
RESOLVED
Raw Incident Data
Source IP
192.168.1.50
Destination IP
N/A
Attack Type
Malware
Failed Attempts
N/A
Target Asset
Database Server
Timestamp
10/5/2026, 1:40:12 PM
AI Reasoning
Classification
Malware Detected
Severity
MEDIUM
Priority
MEDIUM
Priority Score
45 / 100
Recommended Actions
Isolate affected device
Block file hash
Notify administrator
Reasoning Trail (Forward Chaining + Heuristic Search)
1. Incident reported with attack type "Malware".
2. Classification determined as "Malware Detected" based on matching attack-type rules.
3. Target asset "Database Server" matched a high-value keyword (admin/root/database/server) → contributed to severity = "MEDIUM".
4. Heuristic score calculated: severity weight + asset criticality + attack frequency + impact = 45/100 → priority = "MEDIUM".

Project Schedule

Activity	Month 1	Month 2	Month 3	Month 4
Literature Survey	✓			
Problem Identification	✓			
System Design	✓	✓		
Implementation		✓	✓	✓
Testing & Evaluation				✓

Current Progress

- ✅ Project topic and problem identified
- ✅ Literature survey initiated
- ✅ System requirements and methodology finalized
- ✅ Spring Boot project configured
- ✅ MySQL database created
- ✅ Incident entity and database integration completed
- ✅ Repository, Service and Controller implemented
- ✅ CRUD REST APIs developed
- ✅ APIs tested successfully using Postman
- ⬜ AI rule-based analysis implementation
- ⬜ Heuristic prioritization
- ⬜ Response recommendation engine
- ⬜ Frontend dashboard
- ⬜ Final testing and evaluation

Future Work

- Implement the AI-based rule engine.
- Develop Forward Chaining for incident reasoning.
- Implement heuristic-based incident prioritization.
- Develop constraint-based response selection.
- Integrate AI analysis with the Spring Boot backend.
- Develop the cybersecurity monitoring dashboard.
- Perform system testing and evaluate the effectiveness of the proposed approach.
- Complete final documentation and deployment.

Conclusion

- [The proposed AI-Powered Incident Response System aims to automate cybersecurity incident analysis and response recommendation.
- The system combines Cybersecurity, Artificial Intelligence, Backend Development and Database Management.
- AI concepts such as Intelligent Agents, Rule-Based Reasoning, Forward Chaining, Heuristic Evaluation and Constraint Satisfaction are incorporated into the system.
- The system is designed to classify incidents, determine their priority and recommend suitable response actions.
- The final system will provide a centralized dashboard for efficient incident monitoring and AI-assisted decision-making.

Final line:
“The goal is to move from manual incident handling toward intelligent, prioritized and AI-assisted incident response.”

References

[1] Buczak, A. L., & Guven, E., “A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection,” IEEE Communications Surveys & Tutorials, 2016.

[2] Sommer, R., & Paxson, V., “Outside the Closed World: On Using Machine Learning for Network Intrusion Detection,” IEEE Symposium on Security and Privacy, 2010.

[3] Khraisat, A., Gondal, I., Vamplew, P., & Kamruzzaman, J., “Survey of Intrusion Detection Systems: Techniques, Datasets and Challenges,” Cybersecurity, 2019.

[4] Russell, S. J., & Norvig, P., Artificial Intelligence: A Modern Approach, Pearson.